Nested approximate subgroups

Terence Tao · 3 June 2015

https://terrytao.wordpress.com/2015/06/03/nested-approximate-subgroups/

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Suppose that $A \subset B$ are two subgroups of some ambient group G, with the index $K := [B : A]$ of A in B being finite. Then B is the union of K left cosets of A, thus $B = SA$ for some set $S \subset B$ of cardinality K. The elements s of S are not entirely arbitrary with regards to A. For instance, if A is a *normal* subgroup of B, then for each $s \in S$, the conjugation map $g \mapsto s^{-1}gs$ preserves A. In particular, if we write $A^s := s^{-1}As$ for the conjugate of A by s, then

$$A = A^s.$$

Even if A is not normal in B, it turns out that the conjugation map $g \mapsto s^{-1}gs$ *approximately* preserves A, if K is bounded. To quantify this, let us call two subgroups A, B K*-commensurate* for some $K \geq 1$ if one has

$$[A : A \cap B], [B : A \cap B] \leq K.$$

> **Proposition 1** Let $A \subset B$ be groups, with finite index $K = [B : A]$. Then for every $s \in B$, the groups A and A^s are K-commensurate, in fact
>
> $$[A : A \cap A^s] = [A^s : A \cap A^s] \leq K.$$

Proof: One can partition B into K left translates xA of A, as well as K left translates yA^s of A^s. Combining the partitions, we see that B can be partitioned into at most K^2 non-empty sets of the form $xA \cap yA^s$. Each of these sets is easily seen to be a left translate of the subgroup $A \cap A^s$, thus $[B : A \cap A^s] \leq K^2$. Since

$$[B : A \cap A^s] = [B : A][A : A \cap A^s] = [B : A^s][A^s : A \cap A^s]$$

and $[B : A] = [B : A^s] = K$, we obtain the claim. □

One can replace the inclusion $A \subset B$ by commensurability, at the cost of some worsening of the constants:

> **Corollary 2** Let A, B be K-commensurate subgroups of G. Then for every $s \in B$, the groups A and A^s are K^2-commensurate.

Proof: Applying the previous proposition with A replaced by $A \cap B$, we see that for every $s \in B$, $A \cap B$ and $(A \cap B)^s$ are K-commensurate. Since $A \cap B$ and $(A \cap B)^s$ have index at most K in A and A^s respectively, the claim follows. □

It turns out that a similar phenomenon holds for the more general concept of an *approximate group*, and gives a "classification" of all the approximate groups B containing a given approximate group A as a "bounded index approximate subgroup". Recall that a K-approximate group A in a group G for some $K \geq 1$ is a symmetric subset of G containing the identity, such that the product set $A^2 := \{a_1 a_2 : a_1, a_2 \in A\}$ can be covered by at most K left translates of A (and thus also K right translates, by the symmetry of A). For simplicity we will restrict attention to finite approximate groups A so that we can use their cardinality A as a measure of size. We call finite two approximate groups A, B K*-commensurate* if one has

$$\left|A^2 \cap B^2\right| \geq \frac{1}{K}|A|, \frac{1}{K}|B|;$$

note that this is consistent with the previous notion of commensurability for genuine groups.

Theorem 3 Let G be a group, and let $K_1, K_2, K_3 \geq 1$ be real numbers. Let A be a finite K_1-approximate group, and let B be a symmetric subset of G that contains A.

- (i) If B is a K_2-approximate group with $|B| \leq K_3|A|$, then one has $B \subset SA$ for some set S of cardinality at most $K_1K_2K_3$. Furthermore, for each $s \in S$, the approximate groups A and A^s are $K_1K_2^5K_3$-commensurate.
- (ii) Conversely, if $B \subset SA$ for some set S of cardinality at most K_3, and A and A^s are K_2-commensurate for all $s \in S$, then $|B| \leq K_3|A|$, and B is a $K_1^6K_2K_3^2$-approximate group.

Informally, the assertion that B is an approximate group containing A as a "bounded index approximate subgroup" is equivalent to B being covered by a bounded number of shifts sA of A, where s approximately normalises A^2 in the sense that A^2 and $(A^2)^s$ have large intersection. Thus, to classify all such B, the problem essentially reduces to that of classifying those s that approximately normalise A^2.

To prove the theorem, we recall some standard lemmas from arithmetic combinatorics, which are the foundation stones of the "Ruzsa calculus" that we will use to establish our results:

Lemma 4 (Ruzsa covering lemma) If A and B are finite non-empty subsets of G, then one has $B \subset SAA^{-1}$ for some set $S \subset B$ with cardinality $|S| \leq \frac{|BA|}{|A|}$.

Proof: We take S to be a subset of B with the property that the translates sA, $s \in S$ are disjoint in BA, and such that S is maximal with respect to set inclusion. The required properties of S are then easily verified. □

Lemma 5 (Ruzsa triangle inequality) If A, B, C are finite non-empty subsets of G, then

$$\left|AC^{-1}\right| \leq \left|AB^{-1}\right|\left|BC^{-1}\right|/|B|.$$

Proof: If ac^{-1} is an element of AC^{-1} with $a \in A$ and $c \in C$, then from the identity $ac^{-1} = (ab^{-1})(bc^{-1})$ we see that ac^{-1} can be written as the product of an element of AB^{-1} and an element of BC^{-1} in at least $|B|$ distinct ways. The claim follows. □

Now we can prove (i). By the Ruzsa covering lemma, B can be covered by at most

$$\frac{|BA|}{|A|} \leq \frac{\left|B^2\right|}{|A|} \leq \frac{K_2|B|}{|A|} \leq K_2K_3$$

left-translates of A^2, and hence by at most $K_1K_2K_3$ left-translates of A, thus $B \subset SA$ for some $|S| \leq K_1K_2K_3$. Since sA only intersects B if $s \in BA$, we may assume that

$$S \subset BA \subset B^2$$

and hence for any $s \in S$

$$|A^sA| \leq \left|B^2AB^2A\right| \leq \left|B^6\right|$$

$$\leq K_2^5|B| \leq K_2^5K_3 \mid A \mid .$$

By the Ruzsa covering lemma again, this implies that A^s can be covered by at most $K_2^5K_3$ left-translates of A^2, and hence by at most $K_1K_2^5K_3$ left-translates of A. By the pigeonhole principle, there thus exists a group element g with

$$|A^s \cap gA| \geq \frac{1}{K_1K_2^5K_3} \mid A \mid .$$

Since

$$|A^s \cap gA| \leq \left|(A^s \cap gA)^{-1}(A^s \cap gA)\right|$$

and

$$(A^s \cap gA)^{-1}(A^s \cap gA) \subset A^2 \cap (A^s)^2$$

the claim follows.

Now we prove (ii). Clearly

$$|B| \leq |S||A| \leq K_3 \mid A \mid .$$

Now we control the size of $B^2 A$. We have

$$\left|B^2 A\right| \leq \left|SASA^2\right| \leq K_3^2 \sup_{s \in S} \left|AsA^2\right| = K_3^2 \sup_{s \in S} \left|A^s A^2\right|.$$

From the Ruzsa triangle inequality and symmetry we have

$$\begin{aligned}
\left|A^s A^2\right| &\leq \frac{\left|A^s (A^2 \cap (A^2)^s)\right| \left|(A^2 \cap (A^2)^s) A^2\right|}{|A^2 \cap (A^2)^s|} \\
&\leq \frac{\left|(A^3)^s\right| \left|A^4\right|}{|A|/K_2} \\
&\leq K_2 \frac{\left|A^3\right| \left|A^4\right|}{|A|} \\
&\leq K_1^5 K_2 |A|
\end{aligned}$$

and thus

$$\left|B^2 A\right| \leq K_1^5 K_2 K_3^2 \mid A \mid .$$

By the Ruzsa covering lemma, this implies that B^2 is covered by at most $K_1^5 K_2 K_3^2$ left-translates of A^2, hence by at most $K_1^6 K_2 K_3^2$ left-translates of A. Since $A \subset B$, the claim follows.

We now establish some auxiliary propositions about commensurability of approximate groups. The first claim is that commensurability is approximately transitive:

> **Proposition 6** Let A be a K_1-approximate group, B be a K_2-approximate group, and C be a K_3-approximate group. If A and B are K_4-commensurate, and B and C are K_5-commensurate, then A and C are $K_1^2 K_2^3 K_2^3 K_4 K_5 \max(K_1, K_3)$-commensurate.

Proof: From two applications of the Ruzsa triangle inequality we have

$$\begin{aligned}
|AC| &\leq \frac{\left|A(A^2 \cap B^2)\right| \left|(A^2 \cap B^2)(B^2 \cap C^2)\right| \left|(B^2 \cap C^2) C\right|}{|A^2 \cap B^2||B^2 \cap C^2|} \\
&\leq \frac{\left|A^3\right| \left|B^4\right| \left|C^3\right|}{(|A|/K_4)(|B|/K_5)} \\
&\leq K_4 K_5 \frac{K_1^2 \mid A \mid K_2^3 \mid B \mid K_3^2 |C|}{|A||B|} \\
&= K_1^2 K_2^3 K_3^2 K_4 K_5 \mid C \mid .
\end{aligned}$$

By the Ruzsa covering lemma, we may thus cover A by at most $K_1^2 K_2^3 K_3^2 K_4 K_5$ left-translates of C^2, and hence by $K_1^2 K_2^3 K_3^3 K_4 K_5$ left-translates of C. By the pigeonhole principle, there thus exists a group element g such that

$$|A \cap gC| \geq \frac{1}{K_1^2 K_2^3 K_3^3 K_4 K_5} |A|,$$

and so by arguing as in the proof of part (i) of the theorem we have

$$|A^2 \cap C^2| \geq \frac{1}{K_1^2 K_2^3 K_3^3 K_4 K_5} |A|$$

and similarly

$$|A^2 \cap C^2| \geq \frac{1}{K_1^3 K_2^3 K_3^2 K_4 K_5} |C|$$

and the claim follows. □

The next proposition asserts that the union and (modified) intersection of two commensurate approximate groups is again an approximate group:

> **Proposition 7** Let A be a K_1-approximate group, B be a K_2-approximate group, and suppose that A and B are K_3-commensurate. Then $A \cup B$ is a $K_1 + K_2 + K_1^2 K_2^4 K_3 + K_1^4 K_2^2 K_3$-approximate subgroup, and $A^2 \cap B^2$ is a $K_1^6 K_2^3 K_3$-approximate subgroup.

Using this proposition, one may obtain a variant of the previous theorem where the containment $A \subset B$ is replaced by commensurability; we leave the details to the interested reader.

Proof: We begin with $A \cup B$. Clearly $A \cup B$ is symmetric and contains the identity. We have $(A \cup B)^2 = A^2 \cup AB \cup BA \cup B^2$. The set A^2 is already covered by K_1 left translates of A, and hence of $A \cup B$; similarly B^2 is covered by K_2 left translates of $A \cup B$. As for AB, we observe from the Ruzsa triangle inequality that

$$\begin{aligned} |AB^2| &\leq \frac{|A(A^2 \cap B^2)||(A^2 \cap B^2)B^2|}{|A^2 \cap B^2|} \\ &\leq \frac{|A^3||B^4|}{|A|/K_3} \\ &\leq K_1^2 K_2^3 K_3 |B| \end{aligned}$$

and hence by the Ruzsa covering lemma, AB is covered by at most $K_1^2 K_2^3 K_3$ left translates of B^2, and hence by $K_1^2 K_2^4 K_3$ left translates of B, and hence of $A \cup B$. Similarly BA is covered by at most $K_1^4 K_2^2 K_3$ left translates of B. The claim follows.

Now we consider $A^2 \cap B^2$. Again, this is clearly symmetric and contains the identity. Repeating the previous arguments, we see that A is covered by at most $K_1^2 K_2^3 K_3$ left-translates of B, and hence there exists a group element g with

$$|A \cap gB| \geq \frac{1}{K_1^2 K_2^3 K_3} \mid A \mid .$$

Now observe that

$$|(A^2 \cap B^2)^2 (A \cap gB)| \leq |A^5| \leq K_1^4 |A|$$

and so by the Ruzsa covering lemma, $(A^2 \cap B^2)^2$ can be covered by at most $K_1^6 K_2^3 K_3$ left-translates of $(A \cap gB)(A \cap gB)^{-1}$. But this latter set is (as observed previously) contained in $A^2 \cap B^2$, and the claim follows. □

Comments (5)

A SAGE experiment on Nested Approximate Subgroups | Ashani Dasgupta · 6 June 2015
[...] This concerns a computational experiment (in SAGE platform) inspired by an article by Terence Tao on Nested Approximate Subgroup. [...]



Ashani Dasgupta · 8 June 2015
Professor Tao,

I am trying understand this article. However Ruzsa Calculus and allied topics are unknown to me (I have complete in by undergrad course in mathematics last year).
In my experience, tracking down good introductory book in any topic can be a critical and daunting task. So I thought of asking for your suggestion. Can you advice me to find a book on that topic to understand this article more vividly?
Regards,
Ashani Dasgupta
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Terence Tao · 10 June 2015

You can try my book with Van Vu on these topics: https://terrytao.wordpress.com/books/additive-combinatorics/



Ashani Dasgupta · 10 June 2015

Thanks for the response. I will

A SAGE experiment on Nested Approximate Subgroups – Ashani Dasgupta · 3 December 2022
[…] This concerns a computational experiment (in SAGE platform) inspired by an article by Terence Tao on Nested Approximate Subgroup. […]